

Article abstract

Tissues of a multiple sclerosis plaque in the brachium conjunctivum of the pons known to contain peripheral myelin by light microscopic studies were removed from the paraffin block and processed for electron microscopic studies. The cells related to the peripheral myelin possessed the ultrastructural characteristics of Schwann cells, with basement membranes and associated collagen fibers. No continuity was seen with the peripheral nerve tissues of any cranial nerve. These Schwann cells probably arose within the central nervous tissues by selective maturation of multipotential primitive reticular cells, a phenomenon consistent with the view that Schwann cells are mesenchymal in character.

Schwann cells and regenerated peripheral myelin in multiple sclerosis: An ultrastructural study

JUN OGATA, M.D., and IRWIN FEIGIN, M.D.

In light microscopic studies, the myelin characteristic of the central and peripheral nervous systems is readily distinguished by routine staining techniques.¹ By this means, the presence of regenerated myelin of the peripheral type has been disclosed in the lesions of multiple sclerosis²⁻⁴ and in some other conditions,^{5,6} in which they ensheathed preserved central axons that had been denuded of their original central myelin, or might have ensheathed such central fibers.⁷ The cells associated with the peripheral myelin resemble Schwann cells in nuclear morphology and in the presence of collagen and reticulin fibers around them.

In an electron microscopic study of a multiple sclerosis lesion in the spinal cord, the cells around the regenerated peripheral myelin proved to have the ultrastructural characteristics of Schwann cells.⁴ A direct continuity existed between this regenerated peripheral myelin and the myelin in the adjacent nerve roots, so that these investigators thought likely an origin of the Schwann cells from those in the nerve roots.

The present electron microscopic study confirms that the cells associated with regenerated peripheral myelin in multiple sclerosis and in other circumstances do have the ultrastructural characteristics of Schwann cells. In the present instance, however, they cannot be traced to adjacent peripheral nerve tissue of a cranial or spinal nerve.

Case report. (GA15-70). The patient was a 49-year-old woman known to have had multiple sclerosis for many years. The symptomatology included nystagmus, spastic contractures, and inability to speak; her death was due to pneumonia. The brain weighed 1,000 gm. On section, large numbers of sharply demarcated gray plaques were present in the cerebrum, cerebellum, brain stem, and spinal cord, largely in the white matter. Microscopically, plaques were characterized by a marked loss of myelin, a relative preservation of axons, and astrogliosis. In addition, foci of regeneration of myelin of the peripheral type were present in six of the plaques, particularly numerous in a lesion in the brachium conjunctivum of the pons (figure 1).

Method of study and results. The brain originally had been fixed in 20 percent formalin-1 percent acetic acid, cut, embedded in paraffin, sectioned, and stained by standard techniques for light microscopic studies. The precise location in the brachium conjunctivum of the lesion containing large quantities of regenerated

From The Second Department of Internal Medicine, Kyushu University, Fukuoka City, Japan, and the Department of Pathology, New York University and Bellevue Hospital, New York City.

Received for publication January 8, 1975.

Dr. Feigin's address is Department of Pathology, New York University Medical Center, 550 First Avenue, New York, NY 10016.

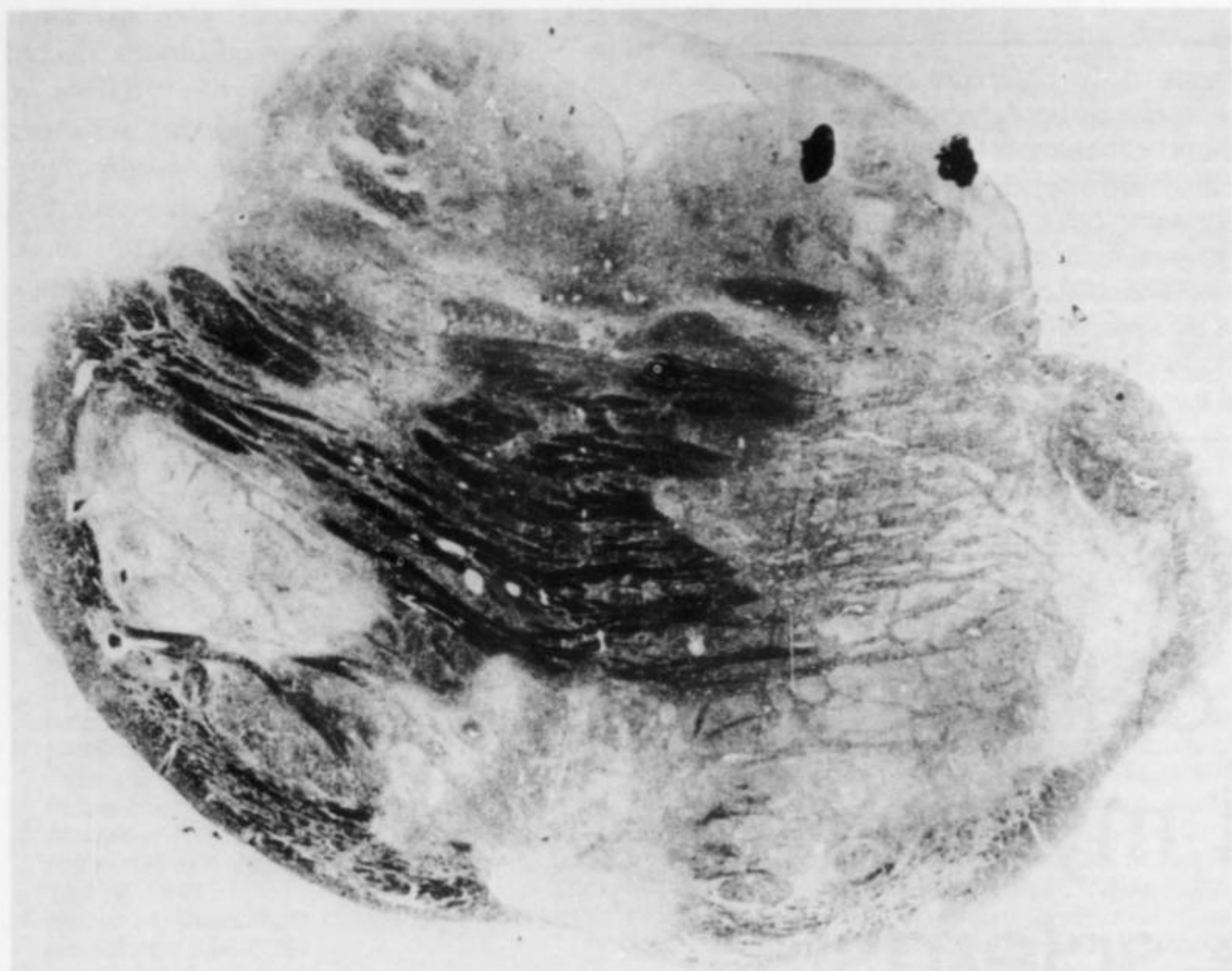


Figure 1. Section of pons. The tissues studied were taken from the area indicated by the ink dots. $\times 4$.

peripheral myelin was identified in the paraffin block by comparison with a stained section. These tissues were excised with a thin knife blade, treated with xylol, graded alcohols and water, postfixed for one hour with 2 percent osmium tetroxide in Millonigs' phosphate buffer,⁸ dehydrated with graded alcohols, treated with propylene oxide, and embedded in epon. Approximately $1\ \mu$ thick sections were prepared, the epon was removed,⁹ and the sections stained with the Luxol blue-periodic acid-Schiff technique in a manner like that used for paraffin sections.¹ These confirmed the presence of myelinated fibers of the peripheral type (staining deep blue), as well as preserved fibers of the central type (staining blue green). Thin sections were cut, doubly stained with uranyl acetate and lead citrate, and examined with the J E M-T8 electron microscope.

This disclosed the presence of myelinated axons, generally more than $1\ \mu$ in diameter. The myelin sheath was surrounded by a cytoplasmic rim, at times with an external mesaxon, beyond which basement membrane material was seen (figures 2, 3). Collagen fibrils showing the characteristic periodicity were present in the extracellular space (figure 4). Astrocytic fibers also were present, and some of these were covered by basement membrane material. No unmyelinated axons were found within Schwann cells.

Discussion. This electron microscopic study confirms the presence of cells with the ultrastructural characteristics of Schwann cells in a central nervous tissue lesion of multiple sclerosis, at a site distant from an exiting peripheral nerve. The nearest such peripheral nerve tissue was at the exit of the fourth nerve seen in an adjacent midbrain section, about which the tissues appeared

normal. In an earlier electron microscopic study, the Schwann cells were traced to adjacent spinal nerves.⁴ In light microscopic studies, it has long been recognized that such Schwann cells could be found at sites, such as the central white matter of the cerebrum, from which a continuity with exiting cranial or spinal nerves could be excluded.^{2,3}

Perivascular peripheral nerves are known to exist within the central nervous tissues, but these too are thought to be an unlikely source of the Schwann cells that produce the peripheral myelin. Perivascular *myelinated* nerves are readily recognized in light microscopic sections by virtue of the specificity and sensitivity of myelin stains;¹ they are rare, present about only a very small proportion of the intracerebral vessels, though they are more common about the extracerebral vessels in the subarachnoid space. When injured, they react by forming myelinated neuromas³ like those seen when peripheral nerves are injured anywhere in the body. These cohesive fibrotic masses containing peripheral myelin are generally distinguishable from the peripheral myelin that ensheaths preserved central axons denuded of their original central myelin in multiple sclerosis. Perivascular *unmyelinated* nerves are very much more difficult to recognize in light microscopic preparations, since their recognition depends on distinguishing axons from connective tissue reticulin fibers in the perivascular tissues, and this cannot be done with certainty with present silver staining techniques. Their incidence can be established indirectly by noting the frequency with which nonmyelinated neuromas¹⁰ are formed in response to nonspecific tissue injury, such as infarcts. These unmyelinated neuromas, and presumably the unmyelinated nerves from which they take origin, are much less frequent than the myelinated nerves and

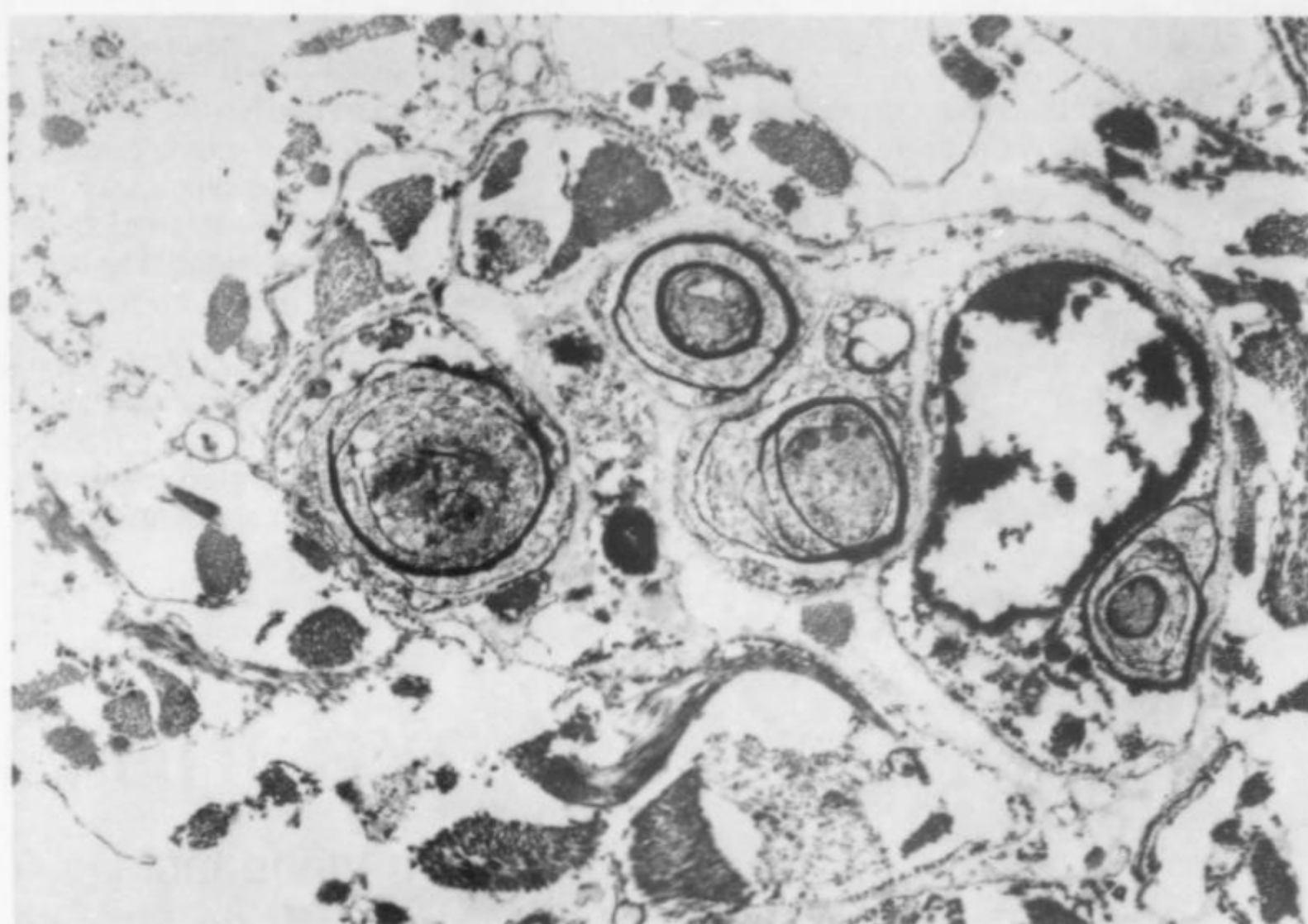


Figure 2. Axons covered by regenerated peripheral myelin are present. The cytoplasm of Schwann cells covered externally by basement membrane material is evident. An external mesaxon is evident on the left. $\times 7,600$.

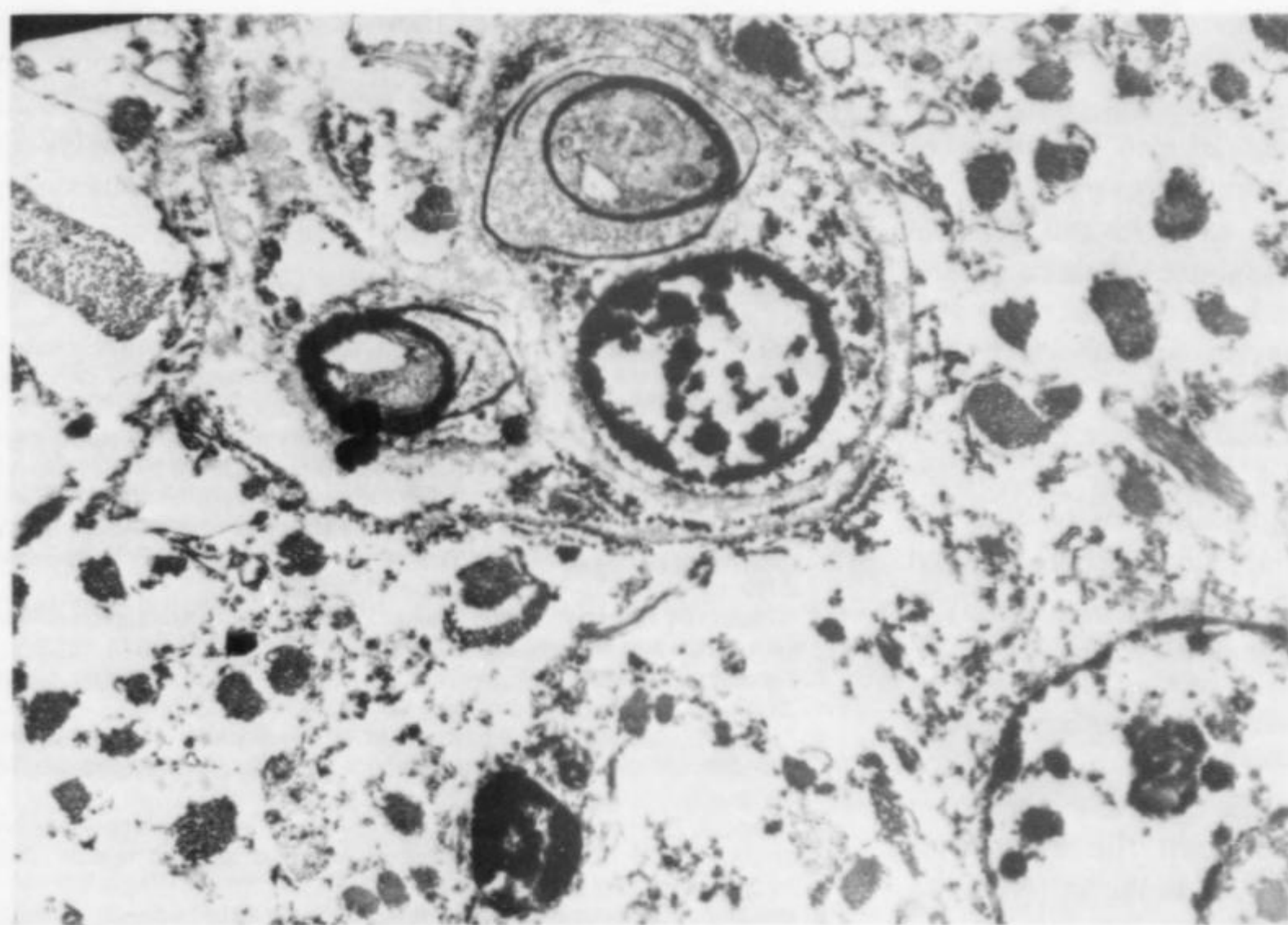


Figure 3. Basement membrane material is evident external to the Schwann cell cytoplasm about the axons covered by regenerated peripheral myelin. $\times 11,000$.

neuromas. The nature of the axon, not differences among the Schwann cell population, determines whether or not an axon will be myelinated;¹¹ the character of the nerve of origin is reflected in the regenerative neuroma that follows its injury, if sufficient time has passed to have permitted myelination, if this were the nature of the nerve involved. The proliferation of Schwann cells within the nonmyelinated neuromas is slight, which further lessens the likelihood of this being the source of the Schwann cells involved in forming the regenerative peripheral myelin in multiple sclerosis. These do not show a centrifugal pattern centered on blood vessels, as might be anticipated if this were their origin.

If Schwann cells are viewed as mesenchymal, they could be formed by maturation of multipotential primitive reticular cells that are present within the central nervous tissues.¹² This would be readily consistent with their ability to form connective tissue fibers and to be transformed into macrophages and would greatly simplify our understanding of the properties of Schwann cells in normal, reactive, and neoplastic circumstances.¹³

The conventional concept that Schwann cells are neuroectodermal is based in part on the intuitive feeling that a cell that produces myelin ought to be neuroectodermal, although other specialized neural elements, such as the leptomeninges and the cerebral blood vessels, which

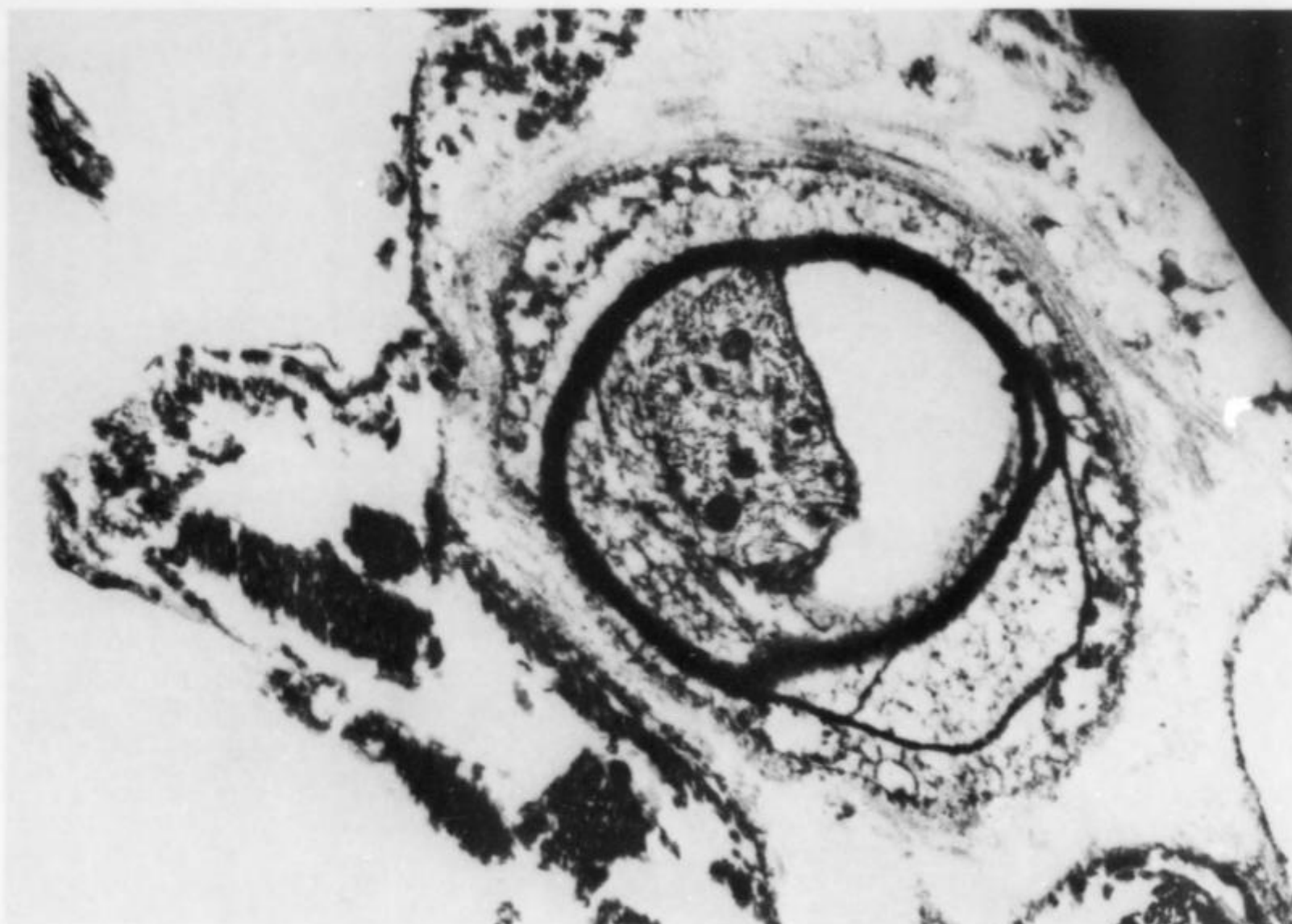


Figure 4. Collagen fibrils are present external to a Schwann cell producing regenerated peripheral myelin about an axon. $\times 14,000$.

differ from those elsewhere, are recognized to be mesenchymal. Their embryologic origin from the neural crest, if this be valid, is also thought to establish a neuroectodermal origin, although it has long been recognized that the neural crest is also a source of mesenchyme.¹⁴ The ultrastructural demonstration of basement membranes in relation to Schwann cells, a feature not shared by fibroblasts and present in the lesion being reported, is an identifying characteristic even when the specific relationship to myelin and axons is not established, as in nerve sheath tumors. Basement membranes are found at the junction between connective tissues and a great variety of epithelial and other "parenchymal" cell elements, including many that are mesenchymal, such as endothelial, mesothelial, fat, and muscle cells, so that the formation of basement membranes is entirely consistent with a mesenchymal origin for Schwann cells.¹⁵

These parenchymal cells are very heterogeneous, and astrocytes are included among them, accounting for the basement membranes between the pia and the most superficial astrocytes, and between the astrocytic processes and the collagenous tissues in the lesion being described in this paper. The basement membranes are thought to be very similar in all circumstances, with only subtle chemical differences observed,¹⁵ so that one might be tempted to implicate the connective tissues always present at one aspect, rather than the markedly heterogeneous parenchymal cells at the other aspect, as the source of the basement membrane material between them, but the experimental data favors the latter nonetheless.¹⁵

The preservation of basement membranes is thought to be essential for an orderly regeneration of many tissues, including that readily noted in the peripheral nervous system, and it has been suggested that the absence of such basement membranes in central nervous tissues accounts for the failure of effective regeneration of axons and myelin sheaths in the brain and spinal cord.¹⁵ However,

the regeneration of peripheral myelin about preserved central axons denuded of their original central myelin in multiple sclerosis,^{2,4} and of peripheral axons and peripheral myelin within the brain following nonspecific injury,^{3,10} suggests that intrinsic differences in the properties of these central and peripheral nervous system elements are involved as well.

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